**Project Title:** Visualizing Housing Market Trends: An Analysis of Sale Prices and Features using Tableau

**Project Design Phase-I - Solution Fit Template**

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Define CS, fit into CC

### 1. CUSTOMER SEGMENT(S)

Who is your customer?

CS

Real estate agents, home buyers, property investors, housing market analysts, and urban planning researchers who need data-driven insights to make informed decisions. This includes first-time home buyers trying to understand fair market pricing, experienced investors comparing property features across price brackets, and real estate agencies looking to benchmark their listings against market averages across 21,609 recorded transactions.

### 6. CUSTOMER CONSTRAINTS

What constraints prevent your customers from taking action or limit their choices of solutions? i.e. spending power, budget, no cash, network connection, available devices.

CC

Customers face limited access to advanced BI tools like Tableau, insufficient technical skills to process large datasets of 21,000+ records, and time pressures that prevent in-depth manual analysis. Most buyer activity is concentrated in the $250K–$450K price range, limiting financial flexibility. Many stakeholders also lack the data literacy to interpret raw outputs, making visual communication essential but often unavailable in traditional tools.

### 5. AVAILABLE SOLUTIONS

Which solutions are available to the customers when they face the problem

AS

Customers rely on Excel for manual analysis, real estate platforms like Zillow for property comparisons, and static PDF reports for summaries. While these tools provide basic information, they lack interactivity, dynamic filtering, and the ability to visualize large datasets comprehensively. None offer a unified view of price distribution, house age, renovation status, and property features together in one place.

Explore AS, differentiate

Focus on J&P, tap into BE, understand RC

### 2. JOBS-TO-BE-DONE / PROBLEMS

Which jobs-to-be-done (or problems) do you address for your customers? There could be more than one; explore different sides.

Buyers, agents, and investors struggle to make sense of thousands of housing records without a clear visual framework. Key challenges include identifying which price brackets dominate the market, understanding how features like bedrooms, bathrooms, and floors relate to sale price, and determining how house age and renovation status influence property value. Without interactive tools, spotting trends quickly or communicating insights to non-technical stakeholders becomes very difficult.

### 9. PROBLEM ROOT CAUSE

RC

What is the real reason that this problem exists? What is the back story behind the need to do this job?
i.e. customers have to do it because of the change in regulations.

The core issue is the disconnect between the volume of available housing data and the tools used to interpret it. The housing market generates complex datasets across thousands of properties with multiple variables — sale price, house age, renovation history, and structural features — that remain hidden without proper visualization tools.
This leads to uninformed pricing decisions, missed investment opportunities, and inefficient market comparisons, ultimately costing buyers, sellers, and investors both time and money.

### 7. BEHAVIOUR

What does your customer do to address the problem and get the job done?

BE

i.e. directly related: find the right solar panel installer, calculate usage and benefits; indirectly associated: customers spend free time on volunteering work (i.e. Greenpeace)

Customers browse real estate websites to compare listings and consult agents for pricing guidance. They attend open houses to physically assess properties and cross-reference multiple listing platforms. Online, they search for average prices filtered by location, house size, and number of bedrooms. Directly related behaviour includes searching for properties within specific price brackets and comparing features like bedrooms, bathrooms, and floors. Associated behaviour includes following housing market news.

Focus on J&P, tap into BE, understand RC

### 3. TRIGGERS

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What triggers customers to act? i.e. seeing their neighbour installing solar panels, reading about a more efficient solution in the news.

Customers are triggered to act when they notice rapidly rising property prices and struggle to determine whether a listing is fairly priced. Reading news reports about housing market shifts and seeing peers make successful property investments motivates others to seek data-driven insights. Upcoming purchase deadlines, mortgage approval windows, and investment decisions further push buyers and agents toward reliable market analysis tools.

### 10. YOUR SOLUTION

SL

If you are working on an existing business, write down your current solution first, fill in the canvas, and check how much it fits reality.
If you are working on a new business proposition, then keep it blank until you fill in the canvas and come up with a solution that fits within customer limitations, solves a problem and matches customer behaviour

. An interactive Tableau dashboard built on 21,609 housing records that makes market trends immediately visible and actionable. The solution includes KPI cards showing an average sale price of $511,619 and total basement area of 38.6M sqft, a bubble chart showing market concentration across price ranges, a bar chart displaying sale price distribution from $50K to $1.1M, a grouped bar chart

### 8. CHANNELS of BEHAVIOUR

CH

**8.1 ONLINE**

What kind of actions do customers take online? Extract online channels from #7

Customers search for properties on real estate websites like Zillow and Realtor.com, filtering by price range, location, and house features. They access interactive Tableau dashboards shared via public links to explore housing market trends. They download digital market reports, compare listings across multiple platforms, and follow housing market news and investment blogs to stay informed about price fluctuations and emerging trends.

**8.2 OFFLINE**

What kind of actions do customers take offline? Extract offline channels from #7 and use them for customer development.

Customers attend open houses to physically assess

Trigger

Fit

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| | **4. EMOTIONS: BEFORE / AFTER** EM<br>How do customers feel when they face a problem or a job and afterwards?<br>i.e. lost, insecure > confident, in control - use it in your communication strategy & design.<br><br>BEFORE: Customers feel overwhelmed and anxious when faced with thousands of raw housing records and no clear way to interpret them. They feel insecure and uncertain when making high-value purchase or investment decisions without solid data backing. Frustration builds when manual comparisons across multiple platforms fail to reveal clear market patterns, leaving them feeling lost and unable to confidently determine whether a property is fairly priced.<br>AFTER: After using the Tableau dashboards, customers feel confident, informed, and in control of their decisions. They experience clarity and relief when complex housing data is transformed into simple, interactive visuals that are easy to understand. Empowered by clear insights into price distributions, house age trends, and renovation impacts, they can make faster, less stressful, and well-supported investment and purchasing decisions. | comparing bedrooms, bathrooms, and floors across different house age groups, and a pie chart breaking down house age distribution by renovation status. This solution fits within customer constraints by replacing complex manual analysis with simple, interactive visuals accessible to non-technical users including agents, buyers, and investors. | properties and consult real estate agents in person for pricing guidance and investment advice. They participate in property expos and community planning sessions to understand broader market trends. Printed dashboard summary reports are shared during agency meetings and investor briefings, allowing non-technical stakeholders to review housing market insights without requiring internet access or digital tools. |